

TECHNICAL REPORT

IEC TR 60825-9

First edition
1999-10

Safety of laser products –

Part 9: Compilation of maximum permissible exposure to incoherent optical radiation

Sécurité des appareils à laser –

Partie 9: Exposition maximale admissible au rayonnement lumineux incohérent



Reference number
IEC/TR 60825-9:1999(E)

Numbering

As from 1 January 1997 all IEC publications are issued with a designation in the 60000 series.

Consolidated publications

Consolidated versions of some IEC publications including amendments are available. For example, edition numbers 1.0, 1.1 and 1.2 refer, respectively, to the base publication, the base publication incorporating amendment 1 and the base publication incorporating amendments 1 and 2.

Validity of this publication

The technical content of IEC publications is kept under constant review by the IEC, thus ensuring that the content reflects current technology.

Information relating to the date of the reconfirmation of the publication is available in the IEC catalogue.

Information on the subjects under consideration and work in progress undertaken by the technical committee which has prepared this publication, as well as the list of publications issued, is to be found at the following IEC sources:

- **IEC web site***
- **Catalogue of IEC publications**
Published yearly with regular updates
(On-line catalogue)*
- **IEC Bulletin**
Available both at the IEC web site* and as a printed periodical

Terminology, graphical and letter symbols

For general terminology, readers are referred to IEC 60050: *International Electrotechnical Vocabulary* (IEV).

For graphical symbols, and letter symbols and signs approved by the IEC for general use, readers are referred to publications IEC 60027: *Letter symbols to be used in electrical technology*, IEC 60417: *Graphical symbols for use on equipment. Index, survey and compilation of the single sheets* and IEC 60617: *Graphical symbols for diagrams*.

* See web site address on title page.

TECHNICAL REPORT

IEC TR 60825-9

First edition
1999-10

Safety of laser products –

Part 9: Compilation of maximum permissible exposure to incoherent optical radiation

Sécurité des appareils à laser –

Partie 9: Exposition maximale admissible au rayonnement lumineux incohérent

© IEC 1999 — Copyright - all rights reserved

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from the publisher.

International Electrotechnical Commission 3, rue de Varembé Geneva, Switzerland
Telefax: +41 22 919 0300 e-mail: inmail@iec.ch IEC web site <http://www.iec.ch>



Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

PRICE CODE

X

For price, see current catalogue

CONTENTS

	Page
FOREWORD	3
Clause	
1 Scope and object.....	4
2 References	5
3 Definitions.....	6
4 Maximum permissible exposure	16
4.1 General remarks.....	16
4.2 Measurement aperture	17
4.3 Pupil diameter.....	18
4.4 Repetitively pulsed, modulated or scanned radiation.....	19
4.5 Angular subtense of the source.....	21
4.6 Time basis.....	23
4.7 Radiance and irradiance	23
4.8 Maximum permissible exposure of the eye.....	24
4.9 Maximum permissible exposure of the skin	34
4.10 Photometric quantities.....	35
5 Measurements.....	35
5.1 Measurement conditions.....	35
5.2 Measurement methods.....	37
Annex A Spectral functions for the Blue-Light-Hazard and the Retinal Thermal Hazard according to ICNIRP	42
Annex B Ultraviolet exposure limits and spectral weighting functions according to ICNIRP	43
Annex C Relative spectral luminous efficiency according to CIE.....	44
Annex D Action spectra.....	45
Annex E Bibliography	49

INTERNATIONAL ELECTROTECHNICAL COMMISSION

SAFETY OF LASER PRODUCTS –**Part 9: Compilation of maximum permissible exposure
to incoherent optical radiation**

FOREWORD

- 1) The IEC (International Electrotechnical Commission) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of the IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, the IEC publishes International Standards. Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. The IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of the IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested National Committees.
- 3) The documents produced have the form of recommendations for international use and are published in the form of standards, technical specifications, technical reports or guides and they are accepted by the National Committees in that sense.
- 4) In order to promote international unification, IEC National Committees undertake to apply IEC International Standards transparently to the maximum extent possible in their national and regional standards. Any divergence between the IEC Standard and the corresponding national or regional standard shall be clearly indicated in the latter.
- 5) The IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with one of its standards.
- 6) Attention is drawn to the possibility that some of the elements of this technical report may be the subject of patent rights. The IEC shall not be held responsible for identifying any or all such patent rights.

The main task of IEC technical committees is to prepare International Standards. However, a technical committee may propose the publication of a technical report when it has collected data of a different kind from that which is normally published as an International Standard, for example "state of the art".

Technical reports do not necessarily have to be reviewed until the data they provide are considered to be no longer valid or useful by the maintenance team.

IEC 60825-9, which is a technical report, has been prepared by IEC technical committee 76: Optical radiation safety and laser equipment.

The text of this technical report is based on the following documents:

Enquiry draft	Report on voting
76/171/CDV	76/204/RVC

Full information on the voting for the approval of this technical report can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 3.

This document which is purely informative is not to be regarded as an International Standard.

SAFETY OF LASER PRODUCTS –

Part 9: Compilation of maximum permissible exposure to incoherent optical radiation

1 Scope and Object

This Technical Report reconciles current **Maximum Permissible Exposure (MPE)** values for the exposure of the human eye and skin to incoherent optical radiation from artificial sources in the wavelength range from 180 nm to 3000 nm with the ultimate goal of harmonisation. Exposure limits between 3000 nm and 1 mm wavelength are currently undefined.

These values are based on the best available information from experimental studies and should be used only as guides in the control of exposure to radiation from artificial sources and should not be regarded as a precise line between safe and dangerous levels.

NOTE The values of this report are applicable to most individuals, however, some individuals may be hypersusceptible or otherwise unusually responsive to optical radiation because of genetic factors, age, personal habits (smoking, alcohol, or other drugs), medication, or previous exposures. Such individuals may not be adequately protected from adverse health effects from exposure to optical radiation at or below the maximum permissible exposure values of this report. Medical advice should be sought to evaluate the extent to which additional protection is needed.

These values were mainly developed for exposure to artificial sources. They may also be used for the evaluation of exposure to sunlight.

The MPE values should not be applicable to exposure of patients to optical radiation for the purpose of medical treatment.

Maximum permissible exposure values for the exposure to radiation from laser sources are defined in IEC 60825-1.

NOTE 1 Basic documents of this report were IEC 60825-1 (addressing lasers) and the IRPA/ICNIRP Guidelines (addressing incoherent sources). ACGIH limits are slightly different in wavelength ranges and in limit values.

NOTE 2 In spite of the fact that LEDs emit mainly incoherent radiation they are currently dealt with in IEC 60825-1.

NOTE 3 There are no damage mechanisms which are known to be different for coherent and incoherent sources. However, in many cases the limit values in IEC 60825-1 are more conservative than the values in this report. This is especially true in wavelength regions where no lasers were available when IEC 60825-1 was originally developed.

NOTE 4 Exposures to levels at the MPE values given may be uncomfortable to view or feel upon the skin.

NOTE 5 In the UV-B and UV-C spectral ranges the MPE values approach the radiant exposures producing minimally detectable biological changes in the surface corneal cells. Levels producing harmful effects are 2 to 3 times greater.

1.1 The object of this technical report is to provide guidance for the protection of persons from incoherent optical radiation in the wavelength range from 180 nm to 1 mm by indicating safe levels of optical radiation which are believed to be safe for most individuals in the sense that exposure at or below these levels will create no adverse effects. Because only limited knowledge exists about the effects of a long-term exposure, most MPEs are based on acute effects of the optical radiation exposure during an eighth hours work day.

1.2 To provide procedures and methods how the level of optical radiation should be measured and evaluated for the purpose of comparison with the maximum permissible exposure.

2 Reference documents

IEC 60050(845):1987, *International Electrotechnical Vocabulary – Chapter 845: Lighting*

IEC 60825-1:1993, *Safety of laser products – Part 1: Equipment classification, requirements and user's guide*
Amendment 1:1997*

ISO 1000:1992, *SI units and recommendations for the use of their multiples and of certain other units*

ISO 11145:1994, *Optics and optical instruments – Lasers and laser-related equipment – Vocabulary and symbols*

ISO/IEC Guide 51:1997, *Safety aspects – Guidelines for their inclusion in standards*

* There is a consolidated edition 1.1 (1998) that includes IEC 60825-1 (1993) and its amendment 1 (1997).

3 Definitions

For the purposes of this report, the following definitions apply. Basic definitions are given in ISO 1000:1992, ISO 11145:1994 and IEC 60050(845):1987. Some of these definitions are repeated, as well as some definitions from IEC 60825-1 and from ISO/IEC Guide 51. Departures from the basic documents are intentional.

(In the following, → means “see”)

3.1

angular magnification M

angular magnification M of an optical instrument is the ratio of the → visual angle subtended by the object with the instrument (α_{instr}) to the visual angle subtended at the eye by the object without the instrument (α_{eye})

$$M = \frac{\alpha_{\text{instr}}}{\alpha_{\text{eye}}}$$

NOTE In technical optics the visual angle subtended without optical instruments is usually based on a comfortable visual distance of 25 cm. In this report the minimum viewing distance is considered to be not smaller than 10 cm.

3.2

angular subtense

visual angle α subtended by the apparent source at the eye of an observer (see figure 1) or at the point of measurement (see also maximum angular subtense and minimum angular subtense)

Symbol: α
SI unit: radian

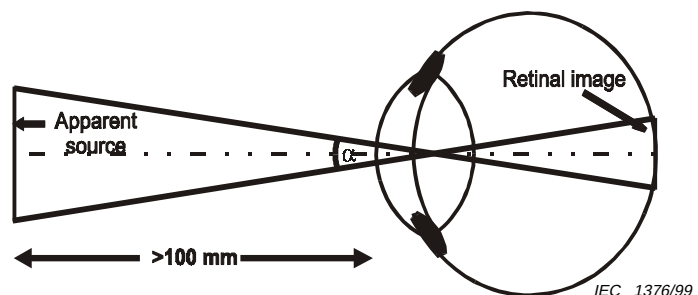


Figure 1 – The definition of the angular subtense α of the apparent source

3.3

aperture, aperture stop

aperture stop is an opening serving to define the area over which radiation is measured (see also measurement aperture)

3.4

apparent source

real or virtual object (source of optical radiation) that forms the smallest possible retinal image

NOTE This definition is used to determine the location of the apparent origin of radiation in the wavelength range of 380 nm to 1400 nm, with the assumption of the apparent source being located in the eye's range of accommodation (usually ≥ 100 mm).

3.5

blue light hazard

potential for a photochemically induced retinal injury resulting from radiation exposure at wavelengths principally between 380 nm and 500 nm

3.6

coherence

characteristic of an electromagnetic field where there is a constant phase relationship between two points in space and time

3.7

coherence length

distance in beam propagation direction within which a constant phase relationship is retained

3.8

diode emitter

any semiconductor p-n junction device which can be made to produce electromagnetic radiation by radiative recombination in the semiconductor in the wavelength range from 180 nm to 1 mm

3.9

exposure distance

shortest distance from a radiation source to the nearest place of human exposure appropriate to the application

3.10

exposure duration

the duration of a pulse, or series, or train of pulses, or of continuous emission of radiation incident upon the human body consistent with the application

3.11

incoherent

radiation is considered incoherent if the coherence length is shorter than 1 mm

3.12

infrared radiation

for practical purposes any electromagnetic radiation within the wavelength range 780 nm to 1 mm. The infrared spectrum is divided into three spectral bands for photobiological safety purposes: →infrared A, →infrared B and →infrared C

3.13

infrared A (IR-A)

optical radiation for which the wavelengths fall in the spectral range from 780 nm to 1400 nm

3.14

infrared B (IR-B)

optical radiation for which the wavelengths fall in the spectral range from 1400 nm to 3000 nm

3.15

infrared C (IR-C)

optical radiation for which the wavelengths fall in the spectral range from 3000 nm to 1 mm

3.16

intended use

the use of a product, process or service in accordance with specifications, instructions and information provided by the supplier

3.17

intermediate source

basically, a source forming an image on the retina which is so large that heat flow in radial direction (perpendicular to the optical axis) from the centre of the image to the surrounding biological tissue is comparable to heat flow in axial directions (parallel to the optical axis)

By extension, an intermediate source is a source forming an image on the retina with a size larger than the size on which the maximum permissible exposure values for →*small sources* are based, up to the size of a →*large source*. This extension is needed because some eye movements are included in the MPEs as listed in the tables of this report

NOTE In this report an intermediate source in its *basic meaning* is subtending at the retina an angle between 1,5 mrad and 100 mrad, i.e. its image diameter on the retina lies between 25 µm and 1700 µm. These dimensions are applicable for exposure times less than 0,7 s.

In this report an intermediate source in its *extended meaning* is subtending at the retina an angle between 11 mrad and 100 mrad, i.e. its image diameter on the retina lies between 187 µm and 1700 µm. These dimensions are valid for exposure times longer than 10 s.

For exposure times between 0,7 s and 10 s the angle subtended by an intermediate source depends on the exposure time (see table 3).

3.18 irradiance

quotient of the →radiant power dP incident on an element of a surface divided by the area dA of that element:

$$E = dP / dA$$

Symbol: E
SI Unit: W / m^2

3.19 large source

source forming an image on the retina which is so large that heat flow in radial direction (perpendicular to the optical axis) from the centre of the image to the surrounding biological tissue is small compared to heat flow in axial directions (parallel to the optical axis)

NOTE In this report a large source is subtending an angle of more than 100 mrad at the retina, i.e. the diameter of its image on the retina is larger than 1700 µm.

3.20 light

→visible radiation

3.21 light emitting diode (LED)

→diode emitter (The optical radiation of LEDs is produced primarily by the process of spontaneous emission)

3.22

maximum angular subtense ($\alpha_{\max}$)

value of angular subtense of the apparent source above which the source is considered to be a →large source (see also table 3)

3.23

maximum permissible exposure (MPE)

value of exposure to the eye or skin which, under normal circumstances, is not expected to result in adverse biological effects. The MPE values are related to the wavelength of the radiation, the exposure duration, the tissue at risk and the dimensions of the exposure site. For visible and near infrared radiation in the range 380 nm to 1 400 nm, the angular subtense of the source defines the size of the retinal image

3.24

measurement aperture

circular area used for measurements of →irradiance, →radiant exposure, →radiance and →time integrated radiance. This aperture defines the area over which these quantities are averaged during measurements for comparison with the MPE values

3.25

monochromatic radiation

radiation characterized by a single wavelength as a single emission line of a low pressure gas discharge lamp. In practice, radiation of a very small wavelength band can be described by stating a single wavelength if the biological action spectrum does not vary significantly within this wavelength band

3.26

optical radiation

electromagnetic radiation at wavelengths between 100 nm and 1 mm. Ultraviolet radiation in the wavelength range below 180 nm (called vacuum UV) is strongly absorbed by the oxygen in the air. For the purpose of this report the wavelength band of optical radiation is limited therefore to wavelengths greater than 180 nm

NOTE Considering the radiation safety, the spectral range between 380 nm and 1400 nm needs special consideration since the eye transmits radiation in this spectral range to the retina, where due to focusing, the irradiance may be increased by several orders of magnitude compared to that at the cornea.

3.27

photometric quantities

all radiometric quantities have corresponding photometric quantities relating to the visual perception of light. For monochromatic radiation of a wavelength λ the photometric quantities can be calculated from the radiometric quantities by multiplying with the relative spectral efficiency $V(\lambda)$ (see annex C) respectively $V'(\lambda)$ and the maximum spectral efficacy of radiation K_m respectively K'_m

$$K_m = 683 \text{ lm / W for photopic vision and}$$

$$K'_m = 1700 \text{ lm / W for scotopic (night) vision}$$

The names of the corresponding radiometric and photometric quantities can be taken from table 1. The symbols are the same for both, if necessary they may be discriminated by the subscript e (energetic) for radiometric and the subscript v (visual) for photometric quantities.

Table 1 – Comparative list of radiometric and photometric quantities

Radiometric quantities		Symbol	Photometric quantities	
Name	Unit		Name	Unit
Radiant power	W	P, Φ	Luminous flux	lm
Radiant energy	J	Q	Quantity of light	lm · s
Irradiance	W/m ²	E	Illuminance	lm/m ² = lx
Radiant exposure	J/m ²	H	Luminous exposure	lx · s
Radiance	W/(sr · m ²)	L	Luminance	lm/(sr · m ²) = cd/m ²
Radiant intensity	W/sr	I	Luminous intensity	lm/sr = cd
Time integrated radiance	J/(sr · m ²)	L_i	Time integrated luminance	lm·s/(sr · m ²)

3.28

pulse duration

the maximum time increment measured between the half peak power points at the leading and trailing edges of a pulse

3.29

radiance

the radiance L in a given direction at a given point is the quotient of the →radiant power dP passing through that point and propagating within the →solid angle $d\Omega$ in a direction ε divided by the product of the area of a section of that beam on a plane perpendicular to this direction ($\cos \varepsilon \cdot dA$) containing the given point and the solid angle $d\Omega$ (see figure 2):

$$L = \frac{dP}{d\Omega \cdot dA \cdot \cos \varepsilon} \quad (1)$$

The same definition holds for the →time integrated radiance L_i if in (1) the radiant power dP is replaced by the radiant energy dQ :

$$L_i = \frac{dQ}{d\Omega \cdot dA \cdot \cos \varepsilon} \quad (2)$$

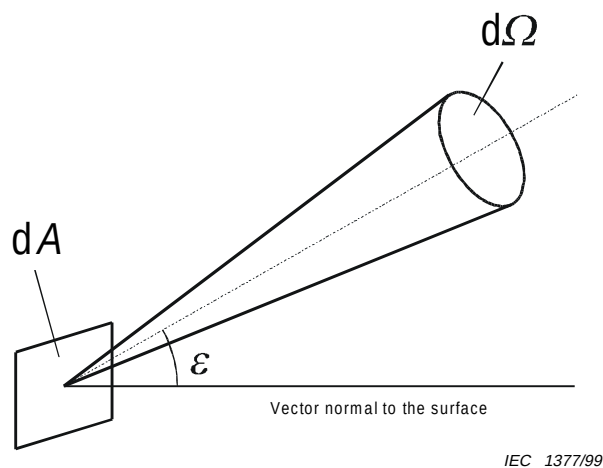
NOTE 1 This definition is a simplified version of IEV 845-01-34, sufficient for the purpose of this report. In cases of doubt, the IEV definition should be followed.

NOTE 2 Radiance and time integrated radiance cannot be changed by optical instruments.

However, if the radiance is measured in a first material with index of refraction n_1 and the radiance is wanted in a second material L_2 with an index of refraction n_2 , the radiance in the first material L_1 has to be multiplied by

the factor $(n_1/n_2)^2$: $\frac{L_2}{L_1} = \left(\frac{n_1}{n_2}\right)^2$. In the case of

air ($n_1 = 1$) and the eye ($n_2 = 1,336$ for aqueous and vitreous) this factor equals to 0,56. For the evaluation of the MPEs the radiance has to be used as measured in air because this factor is already taken into account in the tables of this report.



IEC 1377/99

Figure 2 – Definition of radiance

Symbol of radiance:	L
SI Unit of radiance:	$W / (m^2 \cdot sr)$
Symbol of time integrated radiance:	L_i
SI Unit of time integrated radiance:	$J / (m^2 \cdot sr)$

3.30**radiant energy**

time integral of the →radiant power P over a given duration t

$$Q = \int_0^t P \cdot dt \quad (3)$$

Symbol: Q

SI Unit: Joule (J)

3.31**radiant exposure**

the integral of the →irradiance over a given exposure time, i.e. the ratio of the radiant energy dQ incident on an element of a surface by the area dA of that element:

$$H = dQ / dA \quad (4)$$

Symbol: H

SI Unit: Joule per square metre (J / m²)

3.32**radiant power (flux)**

power emitted, transferred, or received in the form of radiation (IEV 845-01-24)

Symbol: $P, (\Phi)$

SI Unit: Watt (W)

3.33**reflectance**

ratio of the reflected radiant power to the incident radiant power in the given conditions (IEV 845-04-58)

Symbol: ρ

SI Unit: 1

3.34**scanned radiation**

radiation having a time-varying direction, origin or pattern of propagation with respect to a stationary frame of reference

3.35

small source

basically, a source forming an image on the retina which is so small that heat can easily flow in radial direction (perpendicular to the optical axis) from the centre of the image to the surrounding biological tissue

By extension, a source with an image size on the retina smaller than the size on which the maximum permissible exposure values are based. This extension is needed because some eye movements are included in the MPEs as listed in the tables of this report (see also 3.17 and 3.19)

NOTE In this report a small source in its basic meaning is subtending an angle of less than 1,5 mrad at the retina, i.e. the diameter of its image on the retina is less than about 25 µm. This size is applicable for exposure times less than 0,7 s.

A small source in its extended meaning is subtending an angle of less than 11 mrad at the retina, i.e. its diameter is less than about 187 µm. This size is applicable for exposure times longer than 10 s.

For exposure times between 0,7 s and 10 s the limiting angle depends on the exposure time (see table 3).

The term →"point source" cannot be used for small sources because it is prone to confusion: "point sources" may be a lot larger than what is usually considered to be a "point". In this report the term "small source" is therefore used in a similar sense.

3.36

solid angle

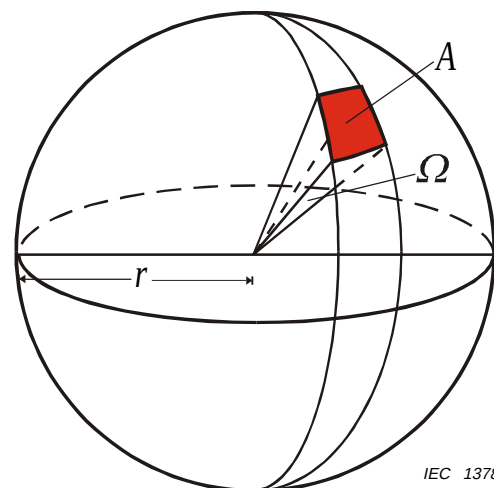
the solid angle with its vertex in the centre of a sphere with the radius r , is the ratio of the area A cut off by this angle of the surface of the sphere divided by the square of the radius (see figure 3):

$$\Omega = A / r^2$$

Symbol: Ω

SI Unit: steradian (sr)

A full solid angle has 4π sr.



IEC 1378/99

Figure 3 – Definition of solid angle

3.37**spectral irradiance**

the ratio of the →radiant power dP in a wavelength interval $d\lambda$ incident on an element of a surface by the area dA of that element and by the wavelength interval $d\lambda$:

$$E_{\lambda} = dP / (dA \cdot d\lambda)$$

Symbol: E_{λ}

SI Unit: W / m^3 , usually expressed as $W / (m^2 \cdot nm)$

3.38**spectral radiance**

the spectral radiance L_{λ} for a wavelength interval $d\lambda$ in a given direction at a given point is the quotient of the →radiant power dP passing through that point and propagating within the →solid angle $d\Omega$ in a direction ε divided by the wavelength interval $d\lambda$ and by the product of the area of a section of that beam on a plane perpendicular to this direction ($\cos \varepsilon \cdot dA$) containing the given point and the solid angle $d\Omega$ (see figure 2):

$$L_{\lambda} = \frac{dP}{d\Omega \cdot d\lambda \cdot dA \cdot \cos \varepsilon}$$

Symbol: L_{λ}

SI Unit: $W / (m^3 \cdot sr)$, usually expressed as $W / (m^2 \cdot sr \cdot nm)$

3.39**time integrated radiance**

the integral of the →radiance over a given exposure time expressed as radiant energy per unit area of a radiating surface per unit solid angle of emission

Symbol: L_i

SI Unit: $J / (m^2 \cdot sr)$

3.40**ultraviolet radiation**

for practical purposes, any radiation within the wavelength range from 100 nm to 400 nm. The ultraviolet spectrum is divided into three spectral bands for photobiological safety purposes: →ultraviolet A, →ultraviolet B and →ultraviolet C. Ultraviolet radiation at wavelengths less than 180 nm is called vacuum ultraviolet radiation

NOTE In many standards the long wavelength limit of the ultraviolet spectral range is fixed at 380 nm.

3.41

ultraviolet A (UV-A)

optical radiation for which the wavelengths fall in the spectral range from 315 nm to 400 nm (see also note above)

3.42

ultraviolet B (UV-B)

optical radiation for which the wavelengths fall in the spectral range from 280 nm to 315 nm

3.43

ultraviolet C (UV-C)

optical radiation for which the wavelengths fall in the spectral range from 100 nm to 280 nm

NOTE Ultraviolet radiation in the wavelength range below 180 nm (called vacuum UV) is strongly absorbed by the oxygen in the air. For the purposes of this report it is therefore sufficient to limit the lower end of the wavelength range of UV-C to 180 nm.

3.44

visible radiation (light)

any optical radiation capable of causing a visual sensation directly (IEV 845-01-03)

NOTE In this report, this is taken to mean electromagnetic radiation for which the wavelength of the monochromatic components lies between 380 nm and 780 nm.

3.45

visual angle

the angle subtended (see 3.2) by an object or detail at the point of observation. It usually is measured in radians, milliradians, degrees or minutes of arc

4 Maximum permissible exposure

4.1 General remarks

Maximum permissible exposure (MPE) values are set below known hazard levels, and are based on the best available information from experimental studies. They apply to exposure within any 8-hour period. The MPE values should be used as guides in the control of exposures and should not be regarded as precisely defined dividing lines between safe and dangerous levels. These limit values do not apply to photosensitive individuals or to individuals concomitantly exposed to photosensitizing agents.

4.2 Measurement aperture

An appropriate aperture has to be used for all measurements and calculations of exposure values. This is the measurement aperture and is defined in terms of the diameter of a circular area over which the irradiance or radiant exposure is to be averaged. Values for these apertures are shown in table 2.

Larger measurement apertures than those given in the table may be used if the irradiance is uniform over the diameter of the measurement aperture and the sensitivity of the detector system makes it necessary. However, with sources of optical radiation that do not produce a uniform optical radiation pattern (i.e. contain hot spots), the apertures according to the table should be used to assess the effect of hot spots.

When evaluating the MPEs of the skin it is recommended to use detectors having a response proportional to the cosine of the angle of incidence of the radiation.

The values of ocular exposures to radiation in the wavelength range from 380 nm to 1400 nm should be measured over a 7 mm diameter aperture (eye pupil).

Table 2 – Minimum aperture diameter applicable to measuring irradiance, radiant exposure, radiance and time integrated radiance

MPEs applicable for the spectral range	See clause	Exposure duration	Diameter of the measure- ment aperture in the case of an exposure of the	
			Eye mm	Skin mm
180 nm to 400 nm	4.8.1	$t \leq 3$ s	1	1
180 nm to 400 nm	4.8.1	$t > 3$ s	7	7
>380 nm to 1400 nm	4.8.2.1, 4.8.2.2	any	7	3,5
>1400 nm to 3000 nm	4.8.2.3	$t \leq 3$ s	1	1
>1400 nm to 3000 nm	4.8.2.3	$t > 3$ s	3,5	3,5

4.3 Pupil diameter

The MPE values applicable to the eye in the wavelength range from 380 nm to 1400 nm given in 4.8.2 are based on a standard pupil diameter d_s of 7 mm for times $<0,5$ s and 3 mm for times $>0,5$ s. Depending on the luminance of the visual field, the diameter of the eye pupil varies between less than 2 mm and more than 7 mm. The pupil diameter varies also from individual to individual, with the visual task, age, etc. The following formula may be used for the calculation of the pupil diameter d_p (measured in mm) from the mean luminance L (measured in cd/m^2) of the object looked at:

$$d_p = 1,29 \text{ mm} + \frac{6,62 \text{ mm}}{1 + \left(\frac{L}{8,24 \text{ cd} / \text{m}^2} \right)^{0,32}} \quad (5)$$

Figure 4 shows this dependence of the pupil diameter on the luminance.

The adjustment of the MPEs in the wavelength range from 380 nm to 1400 nm and for times $< 0,5$ s in relation to the standard pupil diameter d_s (pupil diameter used for MPE specifications) has to be proportional to the area of the pupil:

$$E_{\text{MPE}}(d_p) = E_{\text{MPE}}(d_s) \cdot \left(\frac{d_s}{d_p} \right)^2 \text{ or} \quad (6)$$

$$H_{\text{MPE}}(d_p) = H_{\text{MPE}}(d_s) \cdot \left(\frac{d_s}{d_p} \right)^2 \text{ or} \quad (7)$$

$$L_{\text{MPE}}(d_p) = L_{\text{MPE}}(d_s) \cdot \left(\frac{d_s}{d_p} \right)^2 \text{ resp.} \quad (8)$$

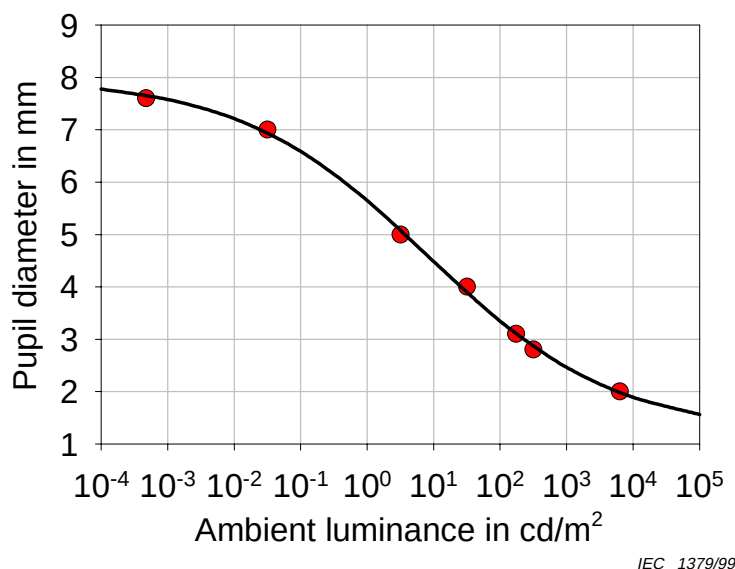


Figure 4 – Dependence of pupil diameter on the luminance of the visual field according to Reeves¹

NOTE In cases where radiation sources are used under very different illumination conditions (e.g. during day, night, etc.), it will be most safe to evaluate the radiation safety for a 7 mm pupil diameter.

4.4 Repetitively pulsed, modulated or scanned radiation

Since there are only limited data on multiple pulse exposure criteria, caution must be used in the evaluation of exposure to repetitively pulsed radiation. Most of the commonly used sources are emitting continuous radiation. However, if the instantaneous value of the radiation output is repetitively falling below 10 % of its time averaged value the following methods should be applied.

For wavelengths <380 nm, the MPE is determined by using the most restrictive of the following requirements a) and b).

The MPE for wavelengths >380 nm is determined by using the most restrictive of requirements b) and c).

- a) The radiant exposure H_{sp} (respectively time integrated radiance L_{sp}) from any single pulse of a duration t within a pulse train should not exceed the MPE H_{MPE} (L_{MPE}) for a single pulse of the duration t :

$$H_{sp} \leq H_{MPE}(t) \quad (9)$$

respectively

¹see e.g. P. Reeves, JOSA 4, 35-43 (1920)

$$L_{sp} \leq L_{MPE}(t) \quad (10)$$

- b) The time averaged irradiance E_m (respectively radiance L_m) for a pulse train of duration T should not exceed the MPE E_{MPE} (respectively L_{MPE}) for a single pulse of duration T .

$$E_m \leq E_{MPE}(T) \quad (11)$$

respectively

$$L_m \leq L_{MPE}(T) \quad (12)$$

The average irradiance E_m (respectively radiance L_m) for the exposure duration T may be calculated by the following relations:

$$E_m = N \cdot H_{sp} / T \quad (13)$$

respectively

$$L_m = N \cdot L_{sp} / T \quad (14)$$

Where N is the total number of pulses during the exposure of duration T .

- c) The radiant exposure H_{sp} (respectively time integrated radiance L_{sp}) from any single pulse of a duration t within a pulse train should not exceed the MPE H_{MPE} (L_{MPE}) for a single pulse of a duration t multiplied by the correction factor C_5 . This correction factor C_5 is only applicable to pulse durations shorter than 0,25 s:

$$H_{sp} \leq H_{MPE}(t) \cdot C_5 \quad (15)$$

respectively

$$L_{sp} \leq L_{MPE}(t) \cdot C_5 \quad (16)$$

where

$$\begin{aligned} C_5 &= N^{-1/4} \\ N &= \text{total number of pulses expected in an exposure.} \end{aligned}$$

These two equations are equivalent to the following equations:

$$\frac{H_{sp}}{H_{MPE}(t) \cdot C_5} = \frac{H_{sp}}{H_{MPE}(t)} \cdot N^{1/4} \leq 1 \quad (17)$$

respectively

$$\frac{L_{sp}}{L_{MPE}(t) \cdot C_5} = \frac{L_{sp}}{L_{MPE}(t)} \cdot N^{1/4} \leq 1 \quad (18)$$

Where a pulse train consists of pulses of different duration t_i or different single pulse radiant exposure H_{spi} (respectively time integrated radiance L_{spi}), the following relations derived from (17) and (18) replace equations (15) and (16):

$$\sum N_i \cdot \left(\frac{H_{spi}}{H_{MPE}(t_i)} \right)^4 \leq 1 \quad (19)$$

respectively

$$\sum N_i \cdot \left(\frac{L_{spi}}{L_{MPE}(t_i)} \right)^4 \leq 1 \quad (20)$$

where

$$\begin{aligned} N_i &= \text{number of pulses of duration } t_i \\ N = \sum N_i &= \text{total number of pulses expected in an exposure.} \end{aligned}$$

In some cases the radiant exposure from a single pulse H_{sp} may fall below the MPE that would apply for continuous exposure at the same peak power using the same total exposure time. Under these circumstances the MPE for continuous exposure may be used.

4.5 Angular subtense of the source

The concept of a limiting →angular subtense of the →apparent source is used in the wavelength range from 380 nm to 1400 nm where the radiation can be focused by the refractive parts of the eye onto the retina.

Two limiting angular subtenses are used in this report: the angle determining the limit between →small sources and →intermediate sources (the minimum angular subtense $\alpha_{\min}$) and the angle determining the limit between intermediate sources and →large sources (the maximum angular subtense $\alpha_{\max}$).

Below the minimum angular subtense ($\alpha_{\min}$) MPEs are independent of the source size. The value of $\alpha_{\min}$ depends on the exposure duration t (see table 3).

NOTE The dependence of the minimum angular subtense on the exposure duration is due to the time dependence of eye movements. For times >10 s the energy will be spread over a larger area on the retina than for times $<0,7$ s. This angle $\alpha_{\min}$ is equal to 11 mrad. For very long exposures of 1000 s and more, when task determined eye movements dominate, the angle is more than 100 mrad.

Table 3 – Limiting angular subtense for the eye

$\alpha_{\min} =$	1,5 mrad	for	$t < 0,7$ s
$\alpha_{\min} =$	$2 \cdot t^{3/4}$ mrad	for	$0,7 \text{ s} \leq t < 10 \text{ s}$
$\alpha_{\min} =$	11 mrad	for	$t \geq 10 \text{ s}$
$\alpha_{\max} =$	100 mrad = 0,1 rad		

Above the maximum angular subtense ($\alpha_{\max}$) MPEs are independent of the source size. The value of $\alpha_{\max}$ does not depend on the exposure time t (see table 3) and is in all cases 100 mrad.

Between the minimum angular subtense ($\alpha_{\min}$) and the maximum angular subtense ($\alpha_{\max}$) MPEs for retinal thermal hazard depend on the source size. Values expressed as radiance or time integrated radiance are inversely proportional to the source size. To describe this dependence of the MPEs on the source size, a correction factor C_{α} is used:

$$\begin{aligned}
 C_{\alpha} &= \alpha_{\min} & \text{for} & & \alpha &\leq & \alpha_{\min} \\
 C_{\alpha} &= \alpha & \text{for} & & \alpha_{\min} < \alpha &\leq & \alpha_{\max} \\
 C_{\alpha} &= \alpha_{\max} & \text{for} & & \alpha_{\max} < \alpha
 \end{aligned}$$

The values for the limiting angular subtense have to be used according to the applicable exposure duration, i.e. $\alpha_{\min} = 1,5$ mrad for single pulses shorter than 0,7 s, and $\alpha_{\min} = 11$ mrad for exposures durations longer than 10 s.

The angular subtense of an oblong source is determined by the arithmetic mean of the maximum and minimum angular dimensions of the source. Any angular dimension that is greater than $\alpha_{\max}$ or less than 1,5 mrad should be limited to $\alpha_{\max}$ or 1,5 mrad respectively, prior to determining the mean.

The angular subtense of the source is determined at the distance of expected exposure. The closest distance at which the human eye can sharply focus is about 100 mm. At shorter distances the image of a light source would be out of focus and blurred. No shorter distance than 100 mm is used therefore in this report for the determination of the angular subtense of a source.

4.6 Time basis

Any evaluation of compliance with the MPEs should be based on the expected exposure duration. When looking into bright sources having a luminance $> 10^4$ cd/m² natural aversion response would limit the exposure to 0,25 s. Where the MPEs expressed in J/m² and exposures longer than 8 hours are expected, there is evidence that under normal extended exposure conditions it is sufficient to integrate in the ultraviolet spectral range the irradiance over 8 hours and to apply the 8-hour MPE.

4.7 Radiance and irradiance

In the following section some MPE values are expressed as radiance (respectively time integrated radiance), some are expressed as irradiance (respectively radiant exposure).

To calculate the irradiance E from the radiance L for an angle $\varepsilon = 0$ (see 3.29) one has to multiply by the solid angle Ω subtended by the source at the eye:

$$E = L \cdot \Omega \quad (21)$$

This relation holds for small solid angles Ω . The more general expression is:

$$dE = L \cdot d\Omega \quad (22)$$

For small circular sources, the following relation holds between the plane angle α and the solid angle Ω :

$$\Omega = \frac{\alpha^2 \cdot \pi}{4} \quad (23)$$

This leads to the following relation between the irradiance and the radiance for a given angular subtense α :

$$E = L \cdot \frac{\alpha^2 \cdot \pi}{4} \quad (24)$$

The equivalent relations hold between time integrated radiance and radiant exposure.

NOTE 1 An instrument measuring radiance normally measures the radiant power through a defined aperture and within a defined angle of acceptance. When applying these relations to the MPEs expressed as radiance in this report, the solid angle Ω of measurement should be calculated using $\alpha_{\min}$.

NOTE 2 When MPEs expressed as radiance are to be measured as irradiance using these relations, the irradiance measurement should be performed with a solid angle Ω corresponding to at least the source size a , but not larger than $(\alpha_{\min}^2 \pi)/4$.

4.8 Maximum permissible exposure of the eye

4.8.1 Ultraviolet spectral range

4.8.1.1 Spectral range between 180 nm and 400 nm

In the spectral range between 180 nm and 400 nm the effective irradiance E_{eff} respectively radiant exposure H_{eff} of a source has to be calculated using the following weighting formula:

$$E_{\text{eff}} = \sum_{180 \text{ nm}}^{400 \text{ nm}} E_{\lambda}(\lambda) \cdot S(\lambda) \cdot \Delta\lambda \quad (25)$$

respectively

$$H_{\text{eff}} = \sum_{180 \text{ nm}}^{400 \text{ nm}} H_{\lambda}(\lambda) \cdot S(\lambda) \cdot \Delta\lambda \quad (26)$$

where $E_{\lambda}(\lambda)$ is the spectral irradiance, $H_{\lambda}(\lambda)$ is the spectral radiant exposure, $S(\lambda)$ is the relative spectral effectiveness (see annex B and figure 5), and $\Delta\lambda$ is the spectral bandwidth.

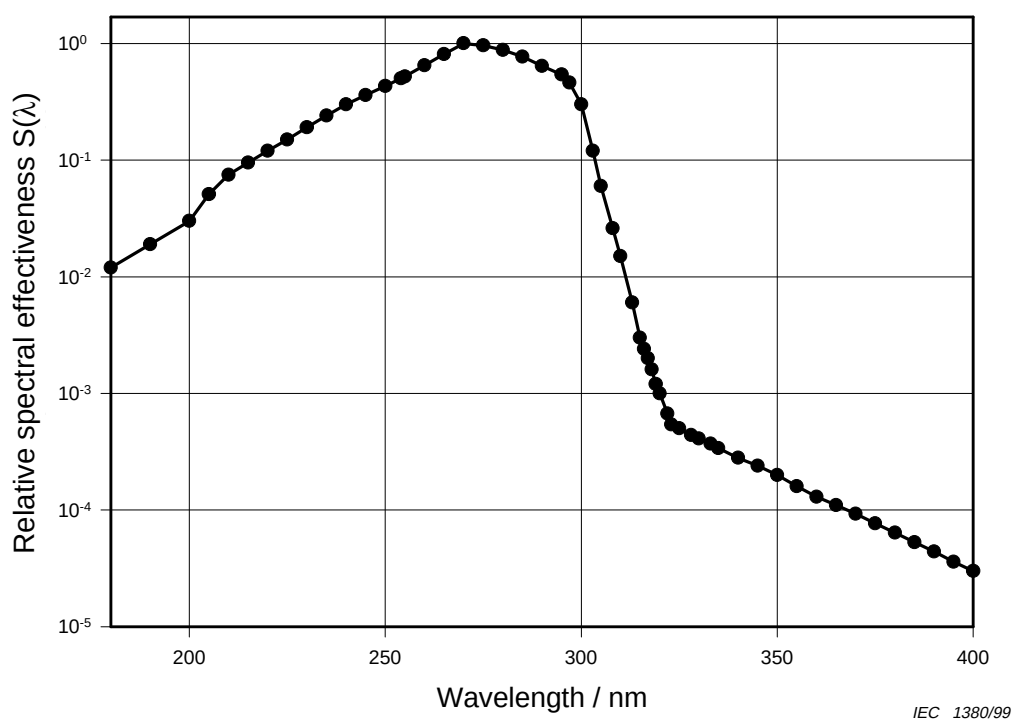


Figure 5 – Relative spectral effectiveness $S(\lambda)$

Table 4 – Maximum Permissible Exposure to UV radiation

Duration of exposure per day	Maximum permissible effective irradiance E_{eff} W/m ²
8 h	0,001
4 h	0,002
2 h	0,004
1 h	0,008
30 min	0,017
15 min	0,033
10 min	0,05
5 min	0,1
30 s	1
10 s	3
1 s	30
0,5 s	60
0,1 s	300

The maximum permissible effective radiant exposure H_{eff} is given by

$$H_{\text{eff}} = 30 \text{ J / m}^2 \quad (27)$$

For a given effective irradiance the permissible exposure time t_{max} in seconds, for exposure to ultraviolet radiation incident upon the unprotected eye is determined by:

$$t_{\text{max}} = \frac{30 \text{ J / m}^2}{E_{\text{eff}}} \quad (28)$$

The exposure time may also be taken from table 4 which provides maximum permissible effective irradiance for a given exposure duration per day.

4.8.1.2 Spectral range between 315 nm and 400 nm

The maximum permissible total radiant exposure within an 8-hour period in the spectral range from 315 nm to 400 nm is

$$H_{\text{UV}} = 10^4 \frac{\text{J}}{\text{m}^2} \quad (29)$$

NOTE In the wavelength range from 315 nm to 400 nm ACGIH extends the 10^4 J/m^2 radiant exposure limit to 1000 s only and specifies for longer times a limit value of 10 W/m^2 .

4.8.2 Visible and infrared spectral ranges

The following three hazard functions should be evaluated: the *Retinal Thermal Hazard*, the *Retinal Blue-Light Photochemical Hazard* and the *Infrared Radiation Hazard* to the cornea and the lens. The most restrictive one of these determines the risk caused by the source.

The maximum permissible exposure values of 4.8.2.1 and 4.8.2.2 are averaged over the standard pupil diameter of 4.3.

4.8.2.1 Retinal Thermal Hazard (380 nm to 1400 nm)

To determine the effective radiance L_{RTH} of a source for the wavelength range from 380 nm to 1400 nm, the following weighting formula should be used:

$$L_{RTH} = \sum_{380 \text{ nm}}^{1400 \text{ nm}} L_{\lambda}(\lambda) \cdot R(\lambda) \cdot \Delta\lambda \quad (30)$$

where $L_{\lambda}(\lambda)$ is the spectral radiance, $R(\lambda)$ is the retinal thermal hazard function (see annex A and figure 6), and $\Delta\lambda$ is the spectral bandwidth.

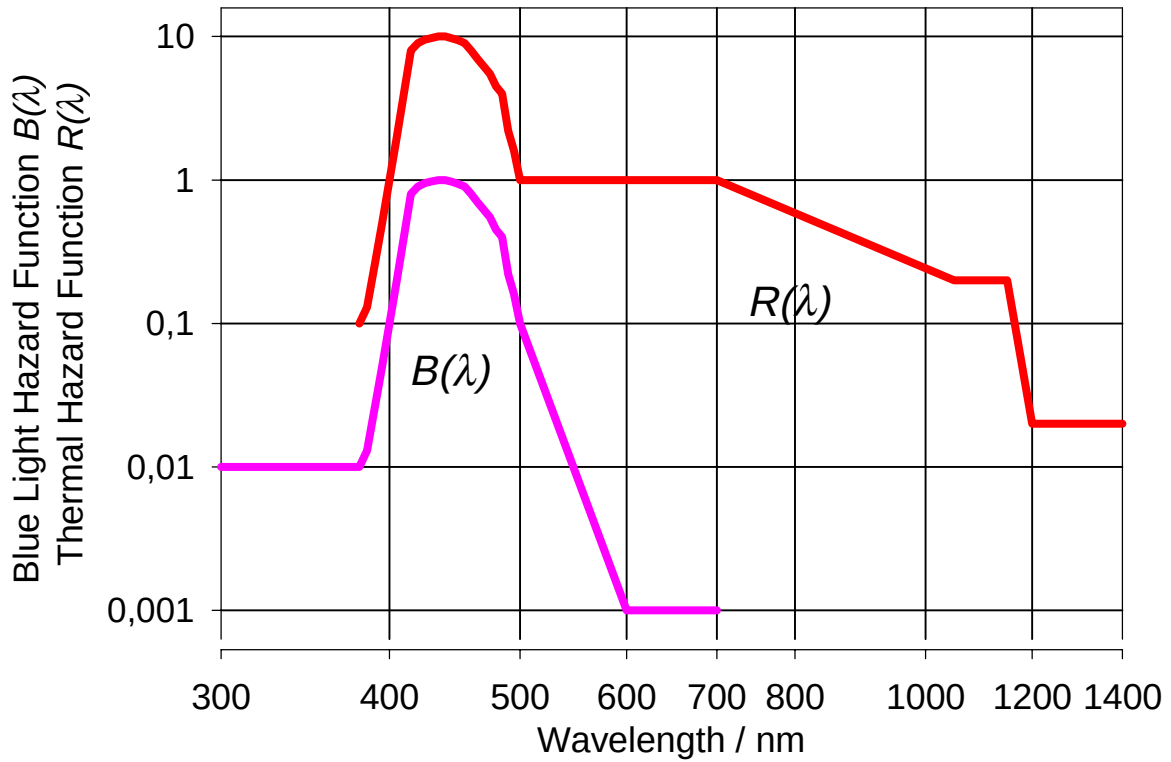
To protect the human retina from thermal injury, the maximum permissible effective radiance L_{RTH} for exposure durations t :

$$10 \text{ s} < t \quad L_{RTH} = \frac{2,8 \cdot 10^4}{C_{\alpha}} \frac{\text{W}}{\text{m}^2 \cdot \text{sr}} \quad (31)$$

$$18 \mu\text{s} \leq t \leq 10 \text{ s} \quad L_{RTH} = \frac{5 \cdot 10^4}{C_{\alpha} \cdot t^{1/4}} \frac{\text{W}}{\text{m}^2 \cdot \text{sr}} \quad (32)$$

$$t < 18 \mu\text{s} \quad L_{RTH} = \frac{41,2}{C_{\alpha} \cdot t^{0,9}} \frac{\text{W}}{\text{m}^2 \cdot \text{sr}} \quad (33)$$

where t is the exposure duration in seconds and C_{α} is the correction factor according to 4.5 in radian.



IEC 1381/99

Figure 6 – Retinal thermal hazard function $R(\lambda)$ and retinal blue light hazard function $B(\lambda)$

The maximum permissible radiance determined using these functions for $\alpha_{\min}$ respectively $\alpha_{\max}$ is shown for wavelengths between 1050 nm and 1150 nm in figure 7.

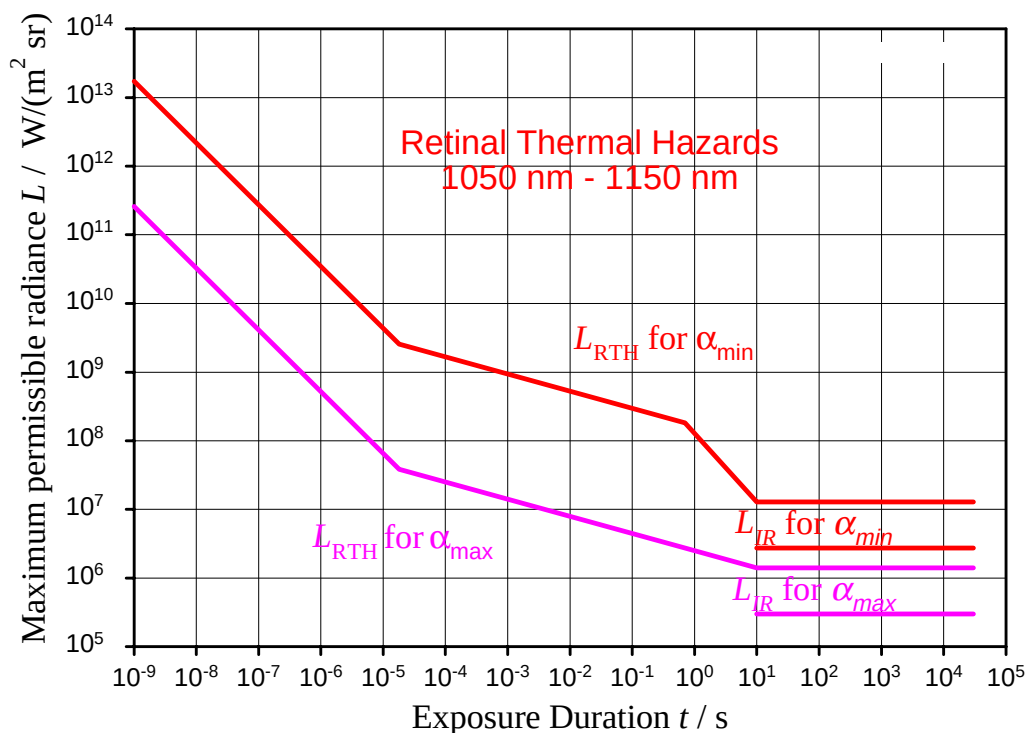
For any source having an angular subtense α greater than 0,1 rad these equations reduce to:

$$10 \text{ s} < t \quad L_{\text{RTH}} = 2,8 \cdot 10^5 \frac{\text{W}}{\text{m}^2 \cdot \text{sr}} \quad (34)$$

$$18 \mu\text{s} \leq t \leq 10 \text{ s} \quad L_{\text{RTH}} = \frac{5 \cdot 10^5}{t^{1/4}} \frac{\text{W}}{\text{m}^2 \cdot \text{sr}} \quad (35)$$

$$t < 18 \mu\text{s} \quad L_{\text{RTH}} = \frac{412}{t^{0,9}} \frac{\text{W}}{\text{m}^2 \cdot \text{sr}} \quad (36)$$

where t is the exposure duration in seconds.



IEC 1382/99

Figure 7 – Time dependence of the maximum permissible radiance L determined by L_{IR} and L_{RTH} using α_{min} respectively α_{max} for the wavelength range from 1050 nm to 1150 nm

For infrared sources with a weak visual stimulus inadequate to activate the aversion response, the effective infrared radiance L_{IR} is determined using the following formula (A weak visual stimulus is defined herein as one whose maximum luminance averaged over a circular field-of-view subtending 0,011 rad is less than $10 \text{ cd}/\text{m}^2$):

$$L_{\text{IR}} = \sum_{780 \text{ nm}}^{1400 \text{ nm}} L_{\lambda}(\lambda) \cdot R(\lambda) \cdot \Delta\lambda \quad (37)$$

where $R(\lambda)$ is the retinal thermal hazard weighting function (see annex A and figure 6) and $L_{\lambda}(\lambda)$ is the spectral radiance of the source.

To protect the human retina from thermal injury the maximum permissible effective infrared radiance L_{IR} should be for:

$$t \geq 10 \text{ s} \quad L_{IR} = \frac{6000}{C_{\alpha}} \frac{W}{m^2 \cdot sr} \quad (38)$$

where C_{α} is the correction factor according to 4.5 in radian.

This function is shown for α_{min} respectively α_{max} and wavelengths between 1050 nm and 1150 nm in figure 7.

For exposure durations shorter than 10 s equations (32) and (33) apply.

4.8.2.2 Retinal Blue-Light Photochemical Hazard (300 nm to 700 nm)

To determine the effective blue-light radiance L_B of a source, the following weighting formula should be used:

$$L_B = \sum_{300 \text{ nm}}^{700 \text{ nm}} L_{\lambda}(\lambda) \cdot B(\lambda) \cdot \Delta\lambda \quad (39)$$

where $L_{\lambda}(\lambda)$ is the spectral radiance and $B(\lambda)$ is the retinal blue-light hazard weighting function (see annex A and figure 6).

To protect the human retina from retinal blue-light injury, the maximum permissible effective radiance L_B should be for:

$$t \leq 10\,000 \text{ s} \quad L_B = \frac{1 \cdot 10^6}{t} \frac{W}{m^2 \cdot sr} \quad (40)$$

where t is the exposure duration in seconds and for

$$t > 10\,000 \text{ s} \quad L_B = 100 \frac{W}{m^2 \cdot sr} \quad (41)$$

NOTE For exposure durations shorter than 10 s, the limit value for the retinal thermal hazard L_{RTH} is usually smaller than the limit value for the blue-light hazard.

When (32) is not satisfied, the maximum direct viewing duration $t_{\max}$ is given by:

$$t_{\max} = \frac{1 \cdot 10^6 \frac{\text{J}}{\text{m}^2 \cdot \text{sr}}}{L_B} \quad (42)$$

To determine the effective blue-light irradiance E_B of a source, the following weighting formula should be used:

$$E_B = \sum_{300 \text{ nm}}^{700 \text{ nm}} E_{\lambda}(\lambda) \cdot B(\lambda) \cdot \Delta\lambda \quad (43)$$

where $E_{\lambda}(\lambda)$ is the spectral irradiance and $B(\lambda)$ is the retinal blue-light hazard weighting function (see Annex A and figure 6).

To protect the human retina from retinal blue-light injury by sources subtending an angle $\alpha < 0,011$ rad the maximum permissible effective blue-light irradiance E_B should be:

$$t \leq 10\,000 \text{ s} \quad E_B = \frac{100 \frac{\text{J}}{\text{m}^2}}{t} \quad (44)$$

$$t > 10\,000 \text{ s} \quad E_B = 0,01 \frac{\text{W}}{\text{m}^2} \quad (45)$$

where t is the exposure duration.

To calculate the maximum direct viewing duration when (44) is not satisfied, this time $t_{\max}$ is given by:

$$t_{\max} = \frac{100 \frac{\text{J}}{\text{m}^2}}{E_B} \quad (46)$$

4.8.2.3 Infrared Radiation Hazard for cornea and lens (780 nm to 3000 nm)

To avoid a thermal hazard to the cornea and possible delayed effects upon the lens of the eye (cataractogenesis), in the spectral range between 780 nm and 3000 nm the total irradiance E_{IR} should be limited for

$$t \geq 1000 \text{ s} \quad E_{\text{IR}} = 100 \frac{\text{W}}{\text{m}^2} \quad (47)$$

$$t < 1000 \text{ s} \quad E_{\text{IR}} = \frac{1,8 \cdot 10^4}{t^{3/4}} \frac{\text{W}}{\text{m}^2} \quad (48)$$

where t is the exposure duration in seconds.

In cold environments, these limits may be increased to 400 W/m² at 0 °C and 300 W/m² at 10 °C for applications where infrared sources are used for radiant heating.

The maximum permissible irradiance values of this section are averaged over the standard pupil diameter of 4.3, however the diameter of the measurement aperture should be taken according to table 2.

4.8.3 Overview on the maximum permissible exposure of the eye

The following table 5 gives an overview on the maximum permissible exposure values for the eye.

Table 5 – Maximum permissible exposure of the eye

Quantity and Weighting Function	Exposure duration <i>t</i> / s Wavelength <i>λ</i> / nm	10-9 to 1,8 x 10-5	1,8 x 10-5 to 10	10 to 103	103 to 104	104 to 3 x 104
$H_{eff} = \sum_{180\text{ nm}}^{400\text{ nm}} H_{\lambda}(\lambda) \cdot S(\lambda) \cdot \Delta\lambda$	180 to 400	30 $\frac{J}{m^2}$				
$H_{UV} = \sum_{315\text{ nm}}^{400\text{ nm}} H_{\lambda}(\lambda) \cdot \Delta\lambda$	315 to 400	$10^4 \frac{J}{m^2}$				
$L_B = \sum_{300\text{ nm}}^{700\text{ nm}} L_{\lambda}(\lambda) \cdot B(\lambda) \cdot \Delta\lambda$	300 to 700	$\frac{1 \cdot 10^6}{t} \frac{W}{m^2 \text{ sr}}$				$100 \frac{W}{m^2 \text{ sr}}$
$L_{RTH} = \sum_{380\text{ nm}}^{1400\text{ nm}} L_{\lambda}(\lambda) \cdot R(\lambda) \cdot \Delta\lambda$	380 to 1400	$\frac{41,2}{C_{\alpha}} \frac{W}{m^2 \text{ sr}}$	$\frac{5 \cdot 10^4}{C_{\alpha}} \frac{W}{m^2 \text{ sr}}$	$\frac{2,8 \cdot 10^4}{C_{\alpha}} \frac{W}{m^2 \text{ sr}}$		
$L_{IR} = \sum_{780\text{ nm}}^{1400\text{ nm}} L_{\lambda}(\lambda) \cdot R(\lambda) \cdot \Delta\lambda$	780 to 1400 Source luminance <10 cd/m²			$\frac{6000}{C_{\alpha}} \frac{W}{m^2 \text{ sr}}$		
$E_{IR} = \sum_{780\text{ nm}}^{3000\text{ nm}} E_{\lambda}(\lambda) \cdot \Delta\lambda$	780 to 3000	$\frac{1,8 \cdot 10^4}{t^{3/4}} \frac{W}{m^2}$			$100 \frac{W}{m^2}$	

4.9 Maximum permissible exposure of the skin

4.9.1 Ultraviolet spectral range (180 nm to 400 nm)

In the spectral range between 180 nm and 400 nm the effective irradiance E_{eff} respectively radiant exposure H_{eff} of a source has to be calculated using the following weighting formula:

$$E_{\text{eff}} = \sum_{180 \text{ nm}}^{400 \text{ nm}} E_{\lambda}(\lambda) \cdot S(\lambda) \cdot \Delta\lambda \quad (49)$$

respectively

$$H_{\text{eff}} = \sum_{180 \text{ nm}}^{400 \text{ nm}} H_{\lambda}(\lambda) \cdot S(\lambda) \cdot \Delta\lambda \quad (50)$$

where $E_{\lambda}(\lambda)$ is the spectral irradiance, $H_{\lambda}(\lambda)$ is the spectral radiant exposure, $S(\lambda)$ is the relative spectral effectiveness (see annex B and figure 5), and $\Delta\lambda$ is the spectral bandwidth.

The maximum permissible effective radiant exposure H_{eff} is given by

$$H_{\text{eff}} = 30 \text{ J / m}^2 \quad (51)$$

For a given effective irradiance the permissible exposure time t_{max} in seconds, for exposure to ultraviolet radiation incident upon the unprotected skin is determined by:

$$t_{\text{max}} = \frac{30 \text{ J / m}^2}{E_{\text{eff}}} \quad (52)$$

The exposure time may also be taken from table 4 which provides exposure times corresponding to a given effective irradiance.

4.9.2 Visible and infrared spectral ranges (380 nm to 3000 nm)

To protect the skin from thermal injury, in the spectral range between 380 nm and 3000 nm the maximum permissible radiant exposure H for exposure durations $t < 10$ s is given by:

$$H = 2 \cdot 10^4 \cdot t^{1/4} \text{ J/m}^2 \quad (53)$$

where t is the exposure duration in seconds. This limit will prevent thermal skin burns.

No limit is provided for exposure durations longer than 10 s because normal avoidance behavior will impose limits on the duration of exposure. Limit values for much longer exposure durations are dominated by concerns of heat stress.

4.10 Photometric quantities

For some radiation sources the emission is characterised by photometric quantities (usually for photopic vision). If the emission is essentially monochromatic, the corresponding radiometric quantity can be calculated by dividing the photometric quantity by the relative spectral efficiency $V(\lambda)$ (see annex C) and the maximum spectral efficacy of radiation K_m (see 3.27). For broadband sources this calculation is not possible if the spectral distribution of the emission of the source is not known.

5 Measurements

5.1 Measurement conditions

Measurements should be made under the following conditions:

5.1.1 At points in space where human access is expected under reasonably foreseeable exposure conditions as determined by a risk assessment.

When evaluating the exposure of the eye, in the wavelength range from 380 nm to 1400 nm, the minimum distance of the measurement aperture from the apparent source should correspond to the minimum distance of intended use. However, it should not be less than 100 mm. The angular subtense α should be determined at the same distance from the apparent source.

5.1.2 With the measuring instrument detector so positioned and so oriented as to result in the maximum detection of radiation by the instrument and averaged over a circular stop according to the measurement aperture applicable according to table 2.

5.1.3 For the evaluation of L_{RTH} and of L_{IR} in the case of apparent sources subtending an angle α greater than α_{min} , a circular measurement aperture having a diameter of 7 mm (see also 4.2 and 4.3) and an effective angle of acceptance equal to α_{min} should be used for evaluating ocular MPEs. In order to detect hot spots, the source should be scanned.

5.1.4 For the evaluation of L_{RTH} and of L_{IR} in the case of apparent sources subtending an angle α less than α_{min} , a circular measurement aperture having a diameter of 7 mm and an effective angle of acceptance equal to α_{min} should be used for evaluating ocular MPEs.

5.1.5 The time dependence of α_{min} needs only to be considered where exposure times between 0,7 s and 10 s become relevant. In most cases it will be sufficient to evaluate a source with the following two angles of acceptance: 1,5 mrad and 11 mrad.

5.1.6 The retinal blue-light hazard should be evaluated with an angle of acceptance equal to α_{min} . Since the retinal blue-light hazard is most restrictive only for longer exposure times (more than about 10 s), in most cases it will be sufficient to evaluate a source with an angle of acceptance of 11 mrad.

5.1.7 In the case where the apparent source consists of multiple points or is a linear source or is a source of non-uniform radiance with an angular subtense greater than α_{min} and within the wavelength range from 380 nm to 1 400 nm, measurements or evaluations of the retinal thermal hazard should be done for every single point, or assembly of points, necessary to assure that the exposure does not exceed the MPE for each possible angle α subtended by each partial area, where $\alpha_{min} \leq \alpha \leq \alpha_{max}$, as well as for the whole source using an angle of acceptance not greater than α_{max} .

5.1.8 For the evaluation of the MPEs, the value of the angular subtense α of a linear source is determined by the mean of the smallest dimension of the source and of the largest dimension of the source. For calculating the mean, the smallest dimension should never been taken smaller than α_{min} , the largest dimension never larger than α_{max} .

5.1.9 MPEs outside the wavelength range from 380 nm to 1400 nm and those applicable for the skin should be evaluated with a detector having a response proportional to the cosine of the angle of incidence of the radiation.

5.2 Measurement methods

The measurement methods described in the following are examples of possible simple methods only. Any other equivalent method may be used. Suitable measurement instruments with known measurement uncertainties are to be used for these measurements.

5.2.1 Measurement of (time integrated) radiance

As given by the definition, the (time integrated) $\rightarrow$ radiance L (L_i) can be determined (see figure 8) by measuring the $\rightarrow$ radiant power P passing through a defined measurement aperture stop having a surface area A at a defined measurement distance. The plane angle of acceptance determines that part of the radiation source over which the radiance is averaged. This plane angle of acceptance α_a defines the $\rightarrow$ solid angle of acceptance Ω of the measurement set-up. For small circular sources, relation (23) between a plane angle α and the solid angle Ω may be used.

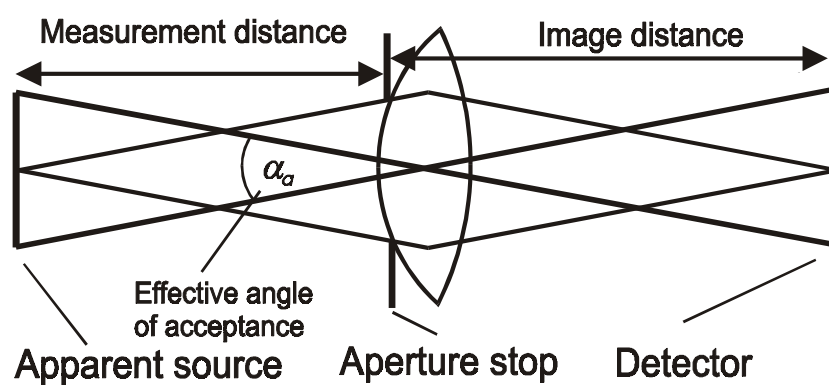


Figure 8 – Measurement conditions for the measurement of (time integrated) radiance

The radiance of the source is calculated from the $\rightarrow$ radiant power P passing through the measurement aperture having a surface area A and the solid angle Ω :

$$L = \frac{P}{\Omega \cdot A} \quad (54)$$

The time integrated radiance L_i is measured in the same way; however, the measurement of the optical power P is replaced by a measurement of the radiant energy Q .

$$L_i = \frac{Q}{\Omega \cdot A} \quad (55)$$

5.2.2 Measurement of the dimensions of the apparent source

5.2.2.1 Bare sources

In many cases, the physical dimensions of the radiating source are known. In these cases the dimensions of the apparent source are equal to the dimensions of the source.

Where the dimensions of the source are not known and the source is not accessible for measurements, the source has to be imaged by e.g. a lens. As an example, such a method is given in figure 9. The lens of focal length f should have a diameter sufficient to intercept the total radiation emitted by the source. Where this is not possible, the minimum diameter d_l of the lens is given by the following relation:

$$d_l = 7 \text{ mm} \cdot \frac{g}{100 \text{ mm}} \quad (56)$$

where g is the distance of the (first principal plane of the) lens from the source. These dimensions are shown in figure 9.

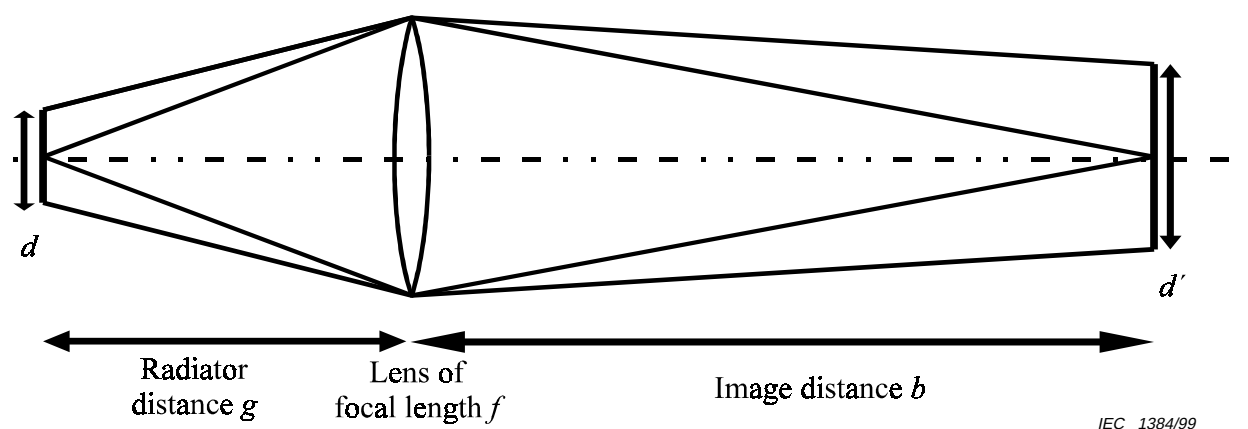


Figure 9 – Determination of the angular subtense of the source

If b is the distance of the image from the (second principal plane of the) lens, then the diameter d of the source is given by the relation

$$d = d' \cdot \frac{g}{b} \quad (57)$$

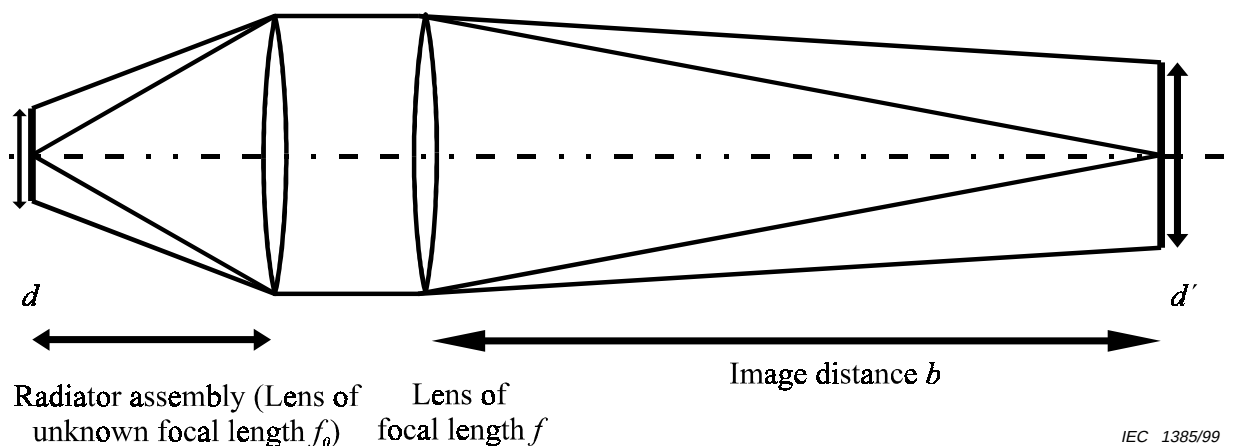
where d' is the image diameter in the image plane. For the determination of d' the methods of 5.2.2.3 and 5.2.2.4 should be used.

5.2.2.2 Source/lens combinations

If the radiator consists of an assembly of a source and an optical element (e.g. a lens and/or a mirror), the distance of the source from the lens, the focal length of the optical element and the positions of the principal planes of the optical element are usually not known. In this case a set-up similar to figure 10 may be used to determine the angular subtense of the source. The diameter of the second lens has to be larger than the diameter of the radiator assembly. For the determination of d' the methods of 5.2.2.3 and 5.2.2.4 should be used.

The magnification M of the source is given by $M = 1 - \frac{a}{f_0} - \frac{b}{f}$ $(1 - \frac{a}{f_0}) - \frac{b}{f}$,

where f_0 is the focal length of the lens of the radiator assembly and a is the distance between the right respectively left principal planes of the two lenses. Determining the image size d' with lenses of different focal lengths f and for several distances a between the two lenses, the two unknown quantities f_0 and d can be evaluated. If a is large enough, it would even not be necessary to determine it separately since the exact position of the principal plane of the unknown lens would scarcely contribute to the measurement error.



IEC 1385/99

Figure 10 – Determination of the angular subtense of the source

5.2.2.3 Sources with circular symmetry

A detector system has to be used with a circular aperture in front of it. Both the detector and the aperture have to be larger than the image of the source to be measured. This assembly is placed in the plane of the image of the source, centred to the optical axis of the propagating beam, and the total power is measured.

The diameter of the aperture is reduced until the power passing through the aperture is equal to 63 % of the total power. The diameter of the aperture at this point gives the diameter of the beam in this plane. The minimum image diameter is measured only, if the aperture is positioned in the plane of the image of the source. It is recommended to repeat the measurement of the beam diameter several times at places close to the assumed position of the image plane.

5.2.2.4 Oblong sources

In the case of rectangular and other oblong sources, the method for the determination of the beam dimensions corresponds to the method for circular sources. However, in place of the circular aperture a rectangular aperture has to be used to determine the smallest possible rectangle through which 63 % of the total power pass. Such noncircular sources have usually two preferential directions to which the sides of the rectangle have to be parallel. The preferential directions are given by the smallest possible beam width and the orthogonal direction. The preferential directions have to be determined at first; this may be done by turning a slit in the plane perpendicular to the plane of propagation of the beam.

To determine the smallest possible rectangle through which 63 % of the total power pass, it is recommended to use an iterative method. To measure the total power, the rectangular aperture in front of the detector has to be large enough not to limit the power passing through it in any direction. Detector and rectangle have to be centred to the optical axis of the propagating beam of radiation.

Now the width of the rectangle has to be reduced in one dimension until $\sqrt{63\%}=79\%$ of the total power pass through the rectangle. In order to check the centring of the rectangular aperture to the beam in the case of sources of non-uniform radiance, the rectangle should be moved back and forward in the direction where the width of the rectangle was varied. This procedure should be repeated until the smallest possible width of the rectangle has been found.

The same procedure has to be repeated for the second preferential dimension of the rectangle.

Both steps should be repeated until the results determined for the width and the length of the image of the source (rectangle) do not change any more.

5.2.3 Determination of the angular subtense

The angular subtense is determined at the distance of intended use respectively expected exposure from the →apparent source. This distance should be not smaller than 100 mm.

The angular subtense of the source is given by the ratio of the source diameter d divided by the viewing distance. The minimum value of the viewing distance is given by the assumed minimum value of the accommodation distance and is equal to 100 mm. In the case of oblong sources two values of angular subtense exist, one for the shortest dimension and a second one for the longest dimension.

In the case of figure 9, the angle α subtended by the image at the lens would be given by

$$\alpha = \arctan (d' / b) \quad (58)$$

This would also be the angle that would be subtended at the eye if the viewing distance would be g . The angle subtended at a different viewing distance g' would be

$$\alpha = \frac{d'}{b} \cdot \frac{g}{g'}. \quad (59)$$

As in the case of figure 9, in the case of figure 10, the angle α subtended by the image at the lens would be given by $\alpha = d' / b$.

Annex A
Spectral functions for the Blue-Light-Hazard and the
Retinal Thermal Hazard according to ICNIRP

Wavelength nm	Blue-Light Hazard Function $B(\lambda)$	Retinal Thermal Hazard function $R(\lambda)$
300-380	0,01	
380	0,01	0,1
385	0,013	0,13
390	0,025	0,25
395	0,05	0,5
400	0,10	1,0
405	0,20	2,0
410	0,40	4,0
415	0,80	8,0
420	0,90	9,0
425	0,95	9,5
430	0,98	9,8
435	1,00	10,0
440	1,00	10,0
445	0,97	9,7
450	0,94	9,4
455	0,90	9,0
460	0,80	8,0
465	0,70	7,0
470	0,62	6,2
475	0,55	5,5
480	0,45	4,5
485	0,40	4,0
490	0,22	2,2
495	0,16	1,6
500	0,10	1,0
>500-600	$10^{0,02 \cdot (450 - \lambda / \text{nm})}$	1
>600-700	0,001	1
>700-1050		$10^{0,002 \cdot (700 - \lambda / \text{nm})}$
>1050-1150		0,2
>1150-1200		$0,2 \cdot 10^{0,02 \cdot (1150 - \lambda / \text{nm})}$
>1200-1400		0,02

Annex B

Ultraviolet exposure limits and spectral weighting functions according to ICNIRP

Wavelength nm	Maximum permissible exposure J/m ²	Relative spectral effectiveness $S(\lambda)$
180	2500	0,012
190	1600	0,019
200	1000	0,030
205	590	0,051
210	400	0,075
215	320	0,095
220	250	0,120
225	200	0,150
230	160	0,190
235	130	0,240
240	100	0,300
245	83	0,360
250	70	0,430
254	60	0,500
255	58	0,520
260	46	0,650
265	37	0,810
270	30	1,000
275	31	0,960
280	34	0,880
285	39	0,770
290	47	0,640
295	56	0,540
297	65	0,460
300	100	0,300
303	250	0,120
305	500	0,060
308	1200	0,026
310	2000	0,015
313	5000	0,0060
315	10000	0,0030
316	13000	0,0024
317	15000	0,0020
318	19000	0,0016
319	25000	0,0012
320	29000	0,0010
322	45000	0,00067
323	56000	0,00054
325	60000	0,00050
328	68000	0,00044
330	73000	0,00041
333	81000	0,00037
335	88000	0,00028
340	110000	0,00028
345	130000	0,00024
350	150000	0,00020
355	190000	0,00016
360	230000	0,00013
365	270000	0,00011
370	320000	0,000093
375	390000	0,000077
380	470000	0,000064
385	570000	0,000053
390	680000	0,000044
395	830000	0,000036
400	1000000	0,000030

Annex C

Relative spectral luminous efficiency according to CIE

Wavelength nm	Relative spectral luminous efficiency for Photopic Vision $V(\lambda)$
380	0,0000
390	0,0001
400	0,0004
410	0,0012
420	0,0040
430	0,0116
440	0,023
450	0,038
460	0,060
470	0,091
480	0,139
490	0,208
500	0,323
510	0,503
520	0,710
530	0,862
540	0,954
550	0,995
560	0,995
570	0,952
580	0,870
590	0,757
600	0,631
610	0,503
620	0,381
630	0,265
640	0,175
650	0,107
660	0,061
670	0,032
680	0,017
690	0,0082
700	0,0041
710	0,0021
720	0,00105
730	0,00052
740	0,00025
750	0,00012
760	0,00006
770	0,00003
780	0,000015

Annex D

Action spectra

The thresholds for the observed bioeffects of UV and infrared radiation at the eye and the skin vary significantly with wavelength. Consequently various "action spectra" have been developed to establish dose-response relationships. In photobiology, the term "action spectrum" refers to the relative spectral effectiveness of different wavelengths in eliciting a biological effect. The MPE values given in this report are "envelopes" over several action spectra. In the UV the action spectrum curve data envelopes e.g. the threshold for single exposure doses (radiant exposures) in the range of reciprocity for acute effects obtained from studies of minimal erythema and keratoconjunctivitis. Reciprocity means that irradiance E and exposure duration t have a reciprocal relation and a constant product of E and t (i.e. radiant exposure H) results in a given effect. Where the biological effects according to the different action spectra need to be evaluated separately, the applicable action spectra have to be used. Some important biological effects of specific wavelength bands are mentioned in the following.

The envelope guideline has some margin of safety to protect all but the most sensitive individuals. An exact value for this margin cannot be given, but for lightly pigmented persons it varies from about 3 to 20 depending on the spectral composition of the radiation.

D.1 Erythema (skin): The threshold for erythema (e.g., the reddening of the skin in sunburn) varies with anatomical site, wavelength, and time between exposure and assessment method. In addition, the variation in threshold values is due to differences in the clinical definition and estimate of minimal erythema and radiometric measurement techniques. Erythema is a photochemical response of the skin normally resulting from overexposure to wavelengths in the UV-C and UV-B regions. The UV-A added prior to UV-B exposure may slightly intensify the erythematous response. This synergistic effect of two spectral bands is known as photoaugmentation.

Action spectra for different grades of erythema are quite different. For the most severe grade of erythema this maximum sensitivity occurs between 290 nm and 300 nm. The minimal erythematous doses (MED) reported ¹⁾, ²⁾, ³⁾ for untanned, lightly pigmented skin range from 60 J/m² to 300 J/m². These MED data suggest that for this type of skin the exposure limit values are approximately 1,3 to 6,5 times less than the MED values. Skin pigmentation and "conditioning" (thickening of the stratum corneum and tanning) may result in an increase of the MED by at least one order of magnitude.

Tentative relative action spectra for the different skin types may be taken from table D.1. When weighted by the relative action spectra the threshold dose for erythema would be 250 J·m⁻². The maximum permissible exposure values in this report are based on the values for sensitive skin.

Table D.1 – Relative action spectra for the formation of erythema

Wavelength range	Relative action spectrum for sensitive skin	Relative action spectrum for moderately sensitive skin
250 nm to 298 nm	1	1
>298 nm to 328 nm	$10^{0,094 \cdot (298 - \lambda / \text{nm})}$	$10^{0,094 \cdot (298 - \lambda / \text{nm})}$
>328 nm to 380 nm	$10^{0,015 \cdot (140 - \lambda / \text{nm})}$	$10^{0,029 \cdot (230 - \lambda / \text{nm})}$

D.2 Pigmentation (skin): UV radiation usually causes pigmentation of the skin (tanning). Pigmentation reduces the sensitivity of the skin to UV radiation by absorption of the radiation in the skin pigment.

Tentative action spectra for the different skin types may be taken from table D.2. When weighted by the relative action spectra the threshold dose for erythema would be 450 J·m⁻².

¹⁾ M. A. Everett, R. L. Olson, R. M. Sayer: Ultraviolet erythema, Arch. Dermatol. **92** (1965), 713-729

²⁾ R. G. Freeman, D. W. Owens, J. M. Knox, H. T. Hudson: Relative energy requirements for an erythematous response of the skin to monochromatic wavelengths for ultraviolet present in the solar spectrum, J. Invest. Dermatol. **47** (1966), 586-592

³⁾ J. A. Parrish, K. F. Jaenicke, R. R. Anderson: Erythema and melanogenesis action spectra of normal human skin, Photochem. Photobiol. **36** (1982), 187-191

Table D.2 – Relative action spectra for pigmentation

Wavelength range	Relative action spectrum for sensitive skin	Relative action spectrum for moderately sensitive skin
250 nm to 298 nm	1	1
>298 nm to 325 nm	$10^{0,094 \cdot (298 - \lambda / \text{nm})}$	$10^{0,094 \cdot (298 - \lambda / \text{nm})}$
>325 nm to 380 nm	$10^{0,015 \cdot (155 - \lambda / \text{nm})}$	$10^{-0,0077 \cdot (4 + \lambda / \text{nm})}$

D.3 Premature skin ageing and skin cancer: Excessive repeated exposures to ultraviolet radiation from the sun or UV appliances may lead to premature ageing of the skin as well as an increased risk for development of skin tumours. The relative effectiveness of different wavelengths to bring about premature skin ageing and skin tumours is presently not known in detail. However, animal studies indicate that the action spectrum for non-melanoma type of skin cancers may be similar to the action spectrum for erythema. Experimental studies on skin of the mouse have shown that UV-B as well as UV-A are complete carcinogens as concerns squamous skin cancer, with the highest effectiveness in the UV-B-region. Several studies in recent years have shown that sunburns, especially in childhood but also in adulthood, raise the risk for malignant melanoma of the skin. Malignant melanoma is the most lethal form of skin cancer. Exact quantitative dose-response relationships for the various forms of skin cancers have not yet been established. Individual variations exist. Certain phenotypes with fair skin, poor ability to develop pigmentation etc. are more prone to develop skin cancers.

D.4 Photokeratoconjunctivitis (eye): Actinic UV radiation (UV-B and UV-C) is strongly absorbed by the cornea and conjunctiva. Overexposure of these tissues causes photokeratoconjunctivitis, commonly referred to as welder's flash, arc-eye, etc. Conjunctivitis tends to develop slowly and may be accompanied by erythema of the facial skin surrounding the eyelids. The individual has the sensation of a foreign body or sand in the eyes and may experience photophobia, lacrimation, and blepharospasm to varying degrees. The acute symptoms last from 6 to 74 hours and discomfort usually disappears within 48 hours. The guideline exposure limits between 180 nm and 305 nm are about 1,3 to 4,6 times less than the threshold for minimal change ⁴⁾.

⁴⁾ D. G. Pitts, T. J. Tredici: The effects of ultraviolet on the eye, Am. Ind. Hyg. Ass. J. **32** (1971), 235-246

The maximum sensitivity of the human eye was found to occur at 270 nm. The wavelength response (relative action spectrum) between 220 nm and 310 nm does not vary as greatly as in the case of erythema with the thresholds varying from 4 J/m² to 140 J/m². Corneal injury from UV-A wavelengths requires levels exceeding 100 J/m².

D.5 Cataract (eye): Wavelengths above 295 nm can be transmitted through the cornea and are absorbed by the lens. Transient and permanent opacities of the lens (cataracts) can be produced in rabbits and monkeys by exposure to UV radiation having wavelengths in the 295 nm to 320 nm band. Thresholds for transient opacities vary dramatically with wavelength ⁵⁾, from 1,5 J/m² to 126 J/m². Thresholds for permanent opacities are typically twice those for transient opacities.

Long term irradiation with high levels of infrared radiation is also known to cause cataract (glass blowers cataract). To avoid possible delayed effects on the lens of the eye, the total irradiance by infrared radiation with wavelengths greater than 780 nm should be limited 100 W/m² for lengthy (>1000 s) exposures. The temperature rise in the eye is responsible for the cataract produced by infrared radiation. Energy absorbed by the cornea, the iris and by the lens may contribute to the temperature rise. It should be limited to less than 1 K. This is achieved by the above mentioned limit and the MPE values in this report.

D.6 Retinal photochemical injury (eye): Exposure to blue light, mainly in the wavelength range below 500 nm, is known to cause a photochemical damage to the retina. This effect is described by the blue light hazard function (see annex A).

⁵⁾ D. G. Pitts, A. P. Cullen, P. D. Hacker: Ocular ultraviolet effects from 295 nm to 400 nm in the rabbit eye, Nat. Inst. Occ. Safety and Health, Cincinnati, OH contract CDC-99-74-12, DHEW (NIOSH) Publ. No. 77-175

Annex E

Bibliography

- ACGIH (American Conference of Governmental Industrial Hygienists) 1997
TLVs and BEIs; Threshold limit values for chemical
substances and physical agents; Biological Exposure
Indices, Cincinnati, ACGIH 1997
- Duchêne A. S. e.a. IRPA (International Radiation Protection Association)
Guidelines on protection against non-ionising radiation,
Pergamon Press, New York 1991
- ICNIRP (International Commission on Non-Ionising Radiation Protection):
Guidelines of limits of exposure to broad-band incoherent
optical radiation (0,38 to 3 μm), Health Physics **73** (1997),
539-554
- IRPA/INIRC (International Non-Ionising Radiation Committee): Guidelines on
limits of exposure to ultraviolet radiation of wavelengths
between 180 nm and 400 nm (Incoherent optical
radiation), Health Physics Vol. **49** (1985), 331-340 and **56**
(1989), 971-972
-



Standards Survey

The IEC would like to offer you the best quality standards possible. To make sure that we continue to meet your needs, your feedback is essential. Would you please take a minute to answer the questions overleaf and fax them to us at +41 22 919 03 00 or mail them to the address below. Thank you!

Customer Service Centre (CSC)

International Electrotechnical Commission

3, rue de Varembé
1211 Genève 20
Switzerland

or

Fax to: **IEC/CSC** at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

A Prioritaire

Nicht frankieren
Ne pas affranchir



Non affrancare
No stamp required

RÉPONSE PAYÉE

SUISSE

Customer Service Centre (CSC)
International Electrotechnical Commission
3, rue de Varembé
1211 GENEVA 20
Switzerland



Q1 Please report on **ONE STANDARD** and **ONE STANDARD ONLY**. Enter the exact number of the standard: (e.g. 60601-1-1)

.....

Q2 Please tell us in what capacity(ies) you bought the standard (tick all that apply). I am the/a:

- purchasing agent ☐
 librarian ☐
 researcher ☐
 design engineer ☐
 safety engineer ☐
 testing engineer ☐
 marketing specialist ☐
 other

Q3 I work for/in/as a:
(tick all that apply)

- manufacturing ☐
 consultant ☐
 government ☐
 test/certification facility ☐
 public utility ☐
 education ☐
 military ☐
 other

Q4 This standard will be used for:
(tick all that apply)

- general reference ☐
 product research ☐
 product design/development ☐
 specifications ☐
 tenders ☐
 quality assessment ☐
 certification ☐
 technical documentation ☐
 thesis ☐
 manufacturing ☐
 other

Q5 This standard meets my needs:
(tick one)

- not at all ☐
 nearly ☐
 fairly well ☐
 exactly ☐

Q6 If you ticked NOT AT ALL in Question 5 the reason is: (tick all that apply)

- standard is out of date ☐
 standard is incomplete ☐
 standard is too academic ☐
 standard is too superficial ☐
 title is misleading ☐
 I made the wrong choice ☐
 other

Q7 Please assess the standard in the following categories, using the numbers:

- (1) unacceptable,
 (2) below average,
 (3) average,
 (4) above average,
 (5) exceptional,
 (6) not applicable

- timeliness
 quality of writing.....
 technical contents.....
 logic of arrangement of contents
 tables, charts, graphs, figures.....
 other

Q8 I read/use the: (tick one)

- French text only ☐
 English text only ☐
 both English and French texts ☐

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

.....



ISBN 2-8318-5024-X



ICS 31.260
